
Summary

Over all 4+ in the end-to-end Data Science Lifecycle, including Problem Identification, Data Collection, Data Cleaning, Statistical Analysis, Data Processing, Exploratory Data Analysis, Feature Engineering, Data Mining, Machine Learning, Predictive modeling, Model training and validation, Hyperparameter Tuning, Model Deployment and monitoring, Data Visualization. Implemented ML governance and lifecycle practices using ML flow, DVC, Airflow, and CI/CD pipelines to ensure reproducibility, monitoring, and compliance across enterprise AI models.

Professional Experience

Bank of America, Charlotte, NC

May 2024 – Present

Data Scientist with AI/ML

- Designed and implemented real-time ML models for credit and debit card fraud detection, applied advanced anomaly detection techniques (Isolation Forest, Autoencoders, One-Class SVM) to uncover suspicious transaction patterns in high-volume data streams.
- Built scalable ETL and streaming pipelines with AWS, Apache Airflow, Spark, and Kafka to process millions of daily transactions.
- Developed feature engineering workflows capturing behavioral patterns (spending velocity, geolocation anomalies, device/browser fingerprints, merchant codes) to improve fraud signal strength.
- Deployed fraud models with containerized microservices (Docker, Kubernetes), ensuring fault-tolerance and scalability for 50K+ transactions per second.
- Applied statistical experiment design (A/B testing, hypothesis testing, uplift modeling) to evaluate new fraud models against baselines, optimizing precision-recall tradeoffs.
- Collaborated with risk management, compliance, and fraud operations teams to ensure alignment with PCI DSS regulations and enterprise fraud prevention strategy.
- Delivered dashboards in Power BI for real-time monitoring of fraud activity, fraud hotspots, and model KPIs (capture rate, false decline rate, throughput).
- Implemented model monitoring pipelines with Prometheus and Jenkins for drift detection, automated retraining, and continuous fraud model performance tracking.
- Explored deployment and experimentation of fraud workflows in Azure ML and Dataiku DSS, ensuring enterprise-readiness and compatibility with hybrid cloud environments.

Tech M, Hyderabad, India 2023

June 2021 – Jul

Data Scientist

- Developed predictive demand forecasting models (LSTM, ARIMA, Gradient Boosting) retail SKUs, optimizing delivery fleet allocation and inventory.
- Designed personalized recommendation systems using collaborative filtering and neural networks, increasing customer engagement and repeat orders.
- Built Databricks- and Spark-based ETL pipelines to integrate and process millions of daily customer, order, and delivery records.
- Applied NLP (Hugging Face, NLTK, Char-CNNs) to analyze customer reviews, support tickets, and delivery feedback, improving customer satisfaction scores by 10% and reducing escalations.
- Modeled cold-chain logistics and last-mile routing optimization for temperature-sensitive food items, lowering spoilage risk and cutting operational cost.
- Developed real-time dashboards in Tableau to monitor demand spikes, fulfillment bottlenecks, delivery SLA adherence, and customer experience KPIs, enabling executives to make faster data-driven decisions.
- Implemented segmentation models (K-means, Hierarchical Clustering) to group customers by order behavior, basket size, and delivery patterns, driving targeted promotions and reducing churn.
- Designed A/B testing frameworks and cross-validation pipelines to evaluate recommendation strategies, demand forecasts, and pricing interventions, ensuring scalability and robustness.
- Established ML monitoring pipelines with ML flow and Airflow for continuous retraining and drift detection, maintaining >93% forecast accuracy during seasonal demand surges.
- Documented system architectures, data workflows, and ML processes to align analytics teams with operations, while mentoring junior data scientists and contributing to best practices.

Zensar Technologies, Hyderabad, India

Sept 2020 – May 2021

Data Analyst with Python

- Designed and implemented ETL processes using Apache Nifi and Talend to integrate financial transaction data, operational metrics, and sales performance, ensuring high-quality data for risk analysis and fraud detection.
- Optimized data pipelines with Apache Spark, reducing processing time by 40% and enabling real-time financial risk insights for Mastercard's payment systems.
- Automated data cleaning and transformation processes using Python and Bash scripting, reducing data preparation time by 35% and improving data quality for accurate risk modeling.
- Developed data warehousing solutions using MySQL and PostgreSQL to store and manage large volumes of financial transaction data, supporting risk analysis and compliance reporting.
- Conducted root cause analysis on financial discrepancies and anomalies using Jupyter Notebook, reducing data anomalies by 30% and ensuring the accuracy of financial risk assessments.
- Implemented continuous integration and deployment workflows using Git and Jenkins, automating the deployment of data pipelines and risk models for seamless updates and real-time insights.
- Enhanced database performance by optimizing SQL queries, enabling faster data retrieval for real-time financial risk monitoring and reporting.
- Leveraged Hadoop and Apache Hive for distributed processing of large-scale financial datasets, improving the efficiency of data storage and processing for Mastercard's global network.

Education

- **Masters in Data Science – University of New Haven - 2024**

Technical Skills

- **Languages & Tools:** Python (pandas, NumPy, matplotlib, seaborn, scikit-learn, TensorFlow, Keras, Hugging Face Transformers, NLTK, Flask, Django), SQL (T-SQL, MySQL, PostgreSQL, Hive), Bash scripting.
- **Cloud Services:** AWS (S3, Redshift, SageMaker, EC2, RDS), Azure ML
- **Data Visualization:** Tableau, Microsoft Power BI, Jupyter Notebook, Pivot Table, matplotlib, seaborn.
- **Machine Learning & AI:** Machine Learning: scikit-learn, TensorFlow, Keras, Random Forest, Gradient Boosting, KNN, ARIMA, Logistic Regression, SVM, NLP
- **DevOps & Collaboration Tools:** Git, Docker, Kubernetes, Jenkins, Hadoop, SAS, MATLAB, Apache Hive, Linux